universidade de aveiro deti departamento de electrónica, telecomunicações e informática

42088 – Industrial Project

Block Diagrams

Milestone 2 – Elaboration

Project name:	Automatic, universal push-button actuator, equipped with force gauge
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1. Introduction

This document presents all the block diagrams in our project divided by levels, each one proving more detailed information on the architecture of the corresponding system.

2. Level 0 Diagram

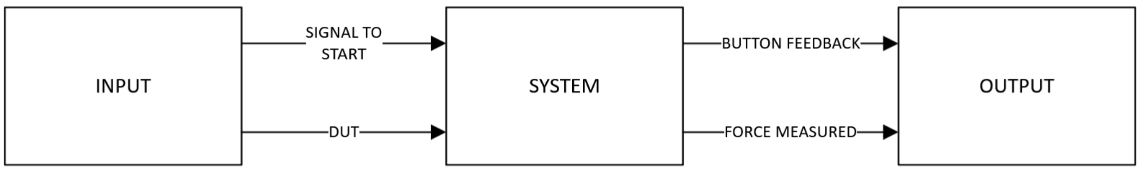

Figure 1. Level 0 Diagram.

The Level 0 diagram of the system represents the simplest view of its functionality. At this level, the system receives two main inputs: a command to start the test and a device under test (DUT) with a specific button to be evaluated. The system then performs the test by pressing the button, and the outputs generated include feedback on the button's operation status and the precise force applied during the press. This high-level view captures the essential workflow, where the system translates a test command into actionable data on button functionality and force measurement.

3. Level 1 Diagram

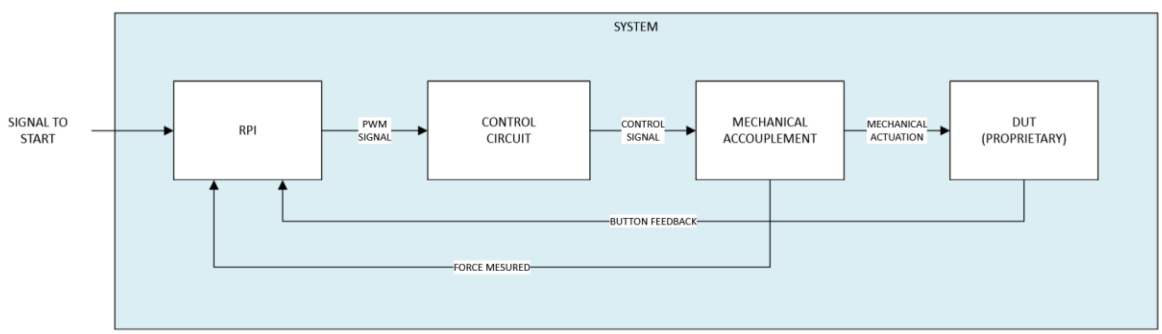

Figure 2. Level 1 Diagram.

The Level 2 diagram of the system provides a closer look at its core components, which are organized into three main blocks:

1. **Raspberry Pi**: Acting as the control unit, the Raspberry Pi serves as the primary interface between the user and the system. It processes user commands, initiates the

test sequence, and communicates with the actuator and force sensor, capturing feedback on button functionality and force data.

2. **Control Circuit**: This circuit is responsible for amplifying the signals output from the Raspberry Pi to match the power requirements of the actuator. It ensures that the actuator receives adequate power and current direction for precise button pressing without straining the Raspberry Pi's limited GPIO output.

3. **Mechanical Actuation**: This component consists of the actuator and force sensor. The actuator is responsible for physically pressing the button on the DUT, while the force sensor captures the exact force applied during each press. Together, they enable accurate and repeatable testing of button functionality.

Each of these blocks is detailed further in subsequent levels, where specific functions, components, and interactions within each part of the system are described.

4. Level 2 Diagrams

4.1 RPI

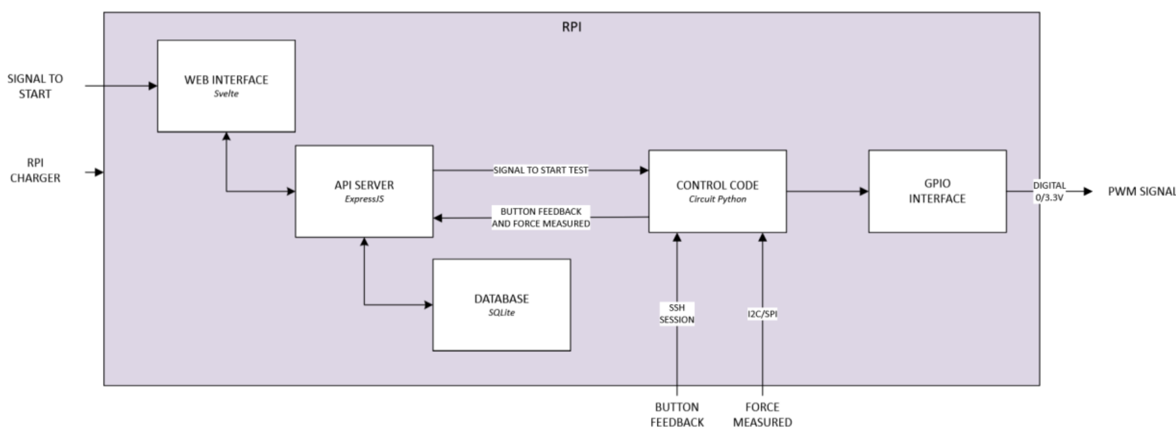

Figure 3. Diagram Level 2 - Raspberry Pi.

In the Level 2 diagram focused on the Raspberry Pi, the system is organized around two primary components: the **web server** and the **control code**.

1. **Web Server**: This component provides the main user interface for configuring and initiating tests. Through the web server, users can interact with the system via a web page accessible from any device within the same network. The web server hosts a database to store test readings, enabling users to access and analyze historical data.

Additionally, it runs an API server that facilitates communication between the web interface, database, and the control code, ensuring seamless data flow and command execution.

2. **Control Code**: This component manages the Raspberry Pi's GPIO interface to control the testing process. It sends a PWM signal to the control circuit to drive the actuator for button pressing. The control code also establishes an SSH session with the DUT, allowing it to read button feedback directly from the device. Additionally, it uses I2C or SPI protocols to communicate with the force sensor, capturing precise force measurements during the test. All readings from the DUT and force sensor are then sent back to the API, making them accessible to the web server for real-time monitoring and record-keeping.

Together, these components allow the Raspberry Pi to control, monitor, and document the testing process, providing a complete interface for remote, automated button testing.

4.2 Control Circuit

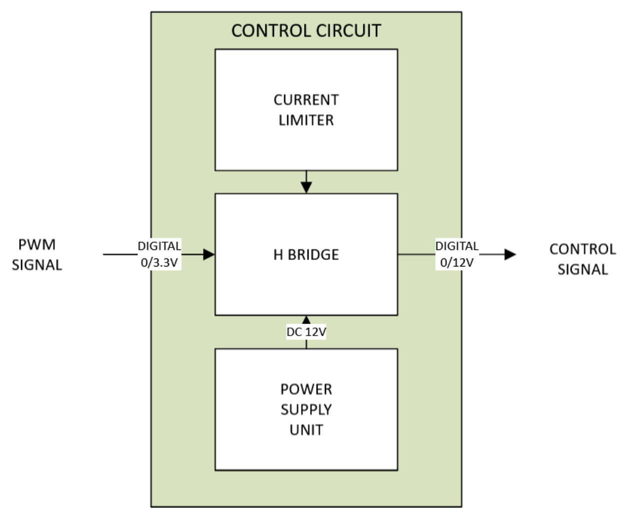

Figure 4. Diagram Level 2 - Control Circuit.

In the Level 2 diagram focused on the **Control Circuit**, the circuit's main purpose is to amplify the output from the Raspberry Pi's GPIO interface to provide sufficient power for the actuator, which the Raspberry Pi alone cannot support.

1. **Signal Amplification**: The Raspberry Pi's GPIO output is too low to directly drive the actuator, so the control circuit amplifies the signal to the necessary power level.

2. **H-Bridge Configuration**: Unlike a standard NOT gate, which only allows current flow in a single direction, the control circuit uses an **H-Bridge** configuration. This setup enables bi-directional current flow, which allows the actuator to both push forward and retract, supporting complete actuation control.

3. **Power Supply**: The entire circuit is powered by a **12V 3A DC power supply**. This supply provides the necessary voltage and current for the actuator to operate smoothly, ensuring stable power delivery throughout the testing process.

Together, these components effectively amplify the PWM signal from the Raspberry Pi and provide controlled, bi-directional movement of the actuator, enabling precise button testing operations.

4.3 Mechanical Accouplement

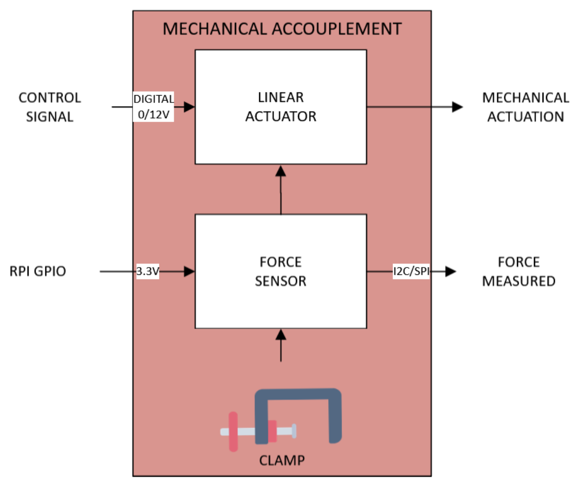

Figure 5. Diagram Level 2 - Mechanical Accouplement.

A clamp secures the force sensor and linear actuator in precise alignment, ensuring that the actuator can press the DUT's button accurately. This design allows for stable positioning and adaptability across various DUTs, keeping the components securely in place during testing.

The force sensor is powered by the Raspberry Pi's GPIO interface, which provides sufficient power for sensor operation. Communication with the force sensor occurs via SPI or I2C protocols to accurately capture force data. The Raspberry Pi's built-in ADC (Analog-to-

Digital Converter) processes these readings, allowing for real-time data capture of the force applied.

The linear actuator is controlled by the amplified PWM signal provided by the control circuit. This amplified signal enables the actuator to apply the required force to press the button, ensuring precise and controlled actuation that matches the DUT's requirements.

4.4 DUT

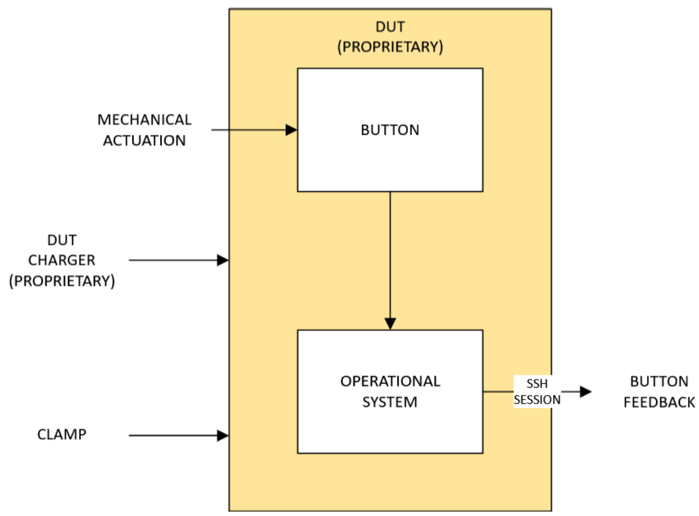

Figure 6. Diagram Level 2 - DUT (Proprietary).

The DUT is a proprietary device, such as a gateway, and is not directly part of our system. However, our system interacts with the DUT in two primary ways to ensure effective testing:

1. **Physical Interaction**: The DUT is securely attached to the clamp within the system. This setup allows the linear actuator to press the DUT's button accurately, ensuring consistent and repeatable tests.

2. **Feedback Monitoring**: During testing, the Raspberry Pi initiates an SSH session with the DUT's operating system, enabling it to monitor real-time feedback. This feedback helps verify button functionality, such as confirming successful button presses or identifying any issues. For this interaction to occur, the DUT must be powered on.

This block illustrates the interface between our system and the DUT, highlighting both the physical alignment for button pressing and the digital connection for monitoring responses.